Product Insights, Observations

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Method 1: Finding Trends in Coupon Usage

Finding Trends in Coupon Usage: Objective

Objective:

Determine how the discount offered contributes to coupon usage.

Factors tested:

- Pizza discount value in comparison to the size of the pizza being discounted.
- Coupons used per week

```
Select CONCAT('https://skynet.indoormedi
From Testimonials t
JOIN FileAssets a on t.ID = a.EntityID
WHERE a.EntityType = 25
ORDER BY t.ID DESC
```

Figure 1: SQL Query

Finding Trends in Coupon Usage: Methodology

- Used SQL Query to extract data from the database. Extracted coupons used/week and image of coupon used.
- Analyzed each individual image and recorded pizza size and discount offered on coupon. Pizza size was standardized by categorizing kids size as 1, small as 2, medium as 3, large as 4, and extra-large as 5.
  - Inch sizes were categorized by assigning a range of inches to each pizza size
- Created a graph based on the recorded information to analyze trends

Finding Trends in Coupon Usage: Example Coupon

- In the Buffo's coupon to the right, we classified the discount as 2$ off of a large pizza, since the pizza size was between 14" and 18".
- Multiple slots were given in case the coupon offered multiple deals. We would then take the average of these discounts

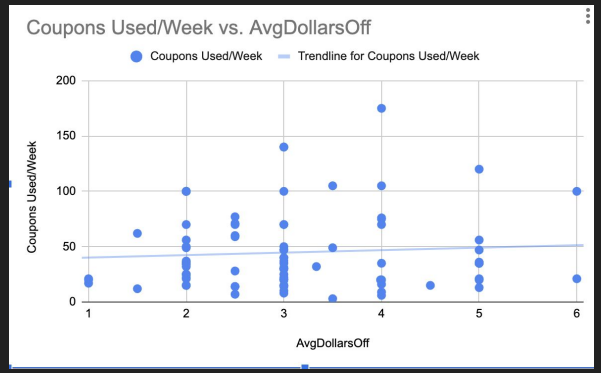
Figure 2: Example coupon and corresponding table entry

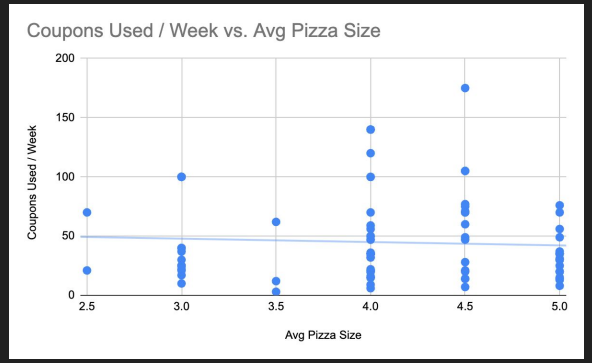
Image Link	Store Name	Comments	Date	PizzaSize1	DollarsOff1	PizzaSize2	DollarsOff2	PizzaSize3	DollarsOff3	Image #
https://skynet.ir	Buffo's	I would recomm	2024-01-10 16:0	4	2					299

Figure 3: example line of data

Finding Trends in Coupon Usage: Findings

- The trendline does not reveal any significant correlation between the average pizza size of the pizza discounted on the coupon and the coupon's usage, nor the size of the pizza and the coupon's usage





Figures 5, 6: Scatterplots showing lack of correlation between average pizza size, discount amount and coupon usage

Finding Trends in Coupon Usage: Possible Inaccuracies

- Four different people went through this data and manually recorded the qualities. Thus, there was a lot of room for human error in the recorded data.

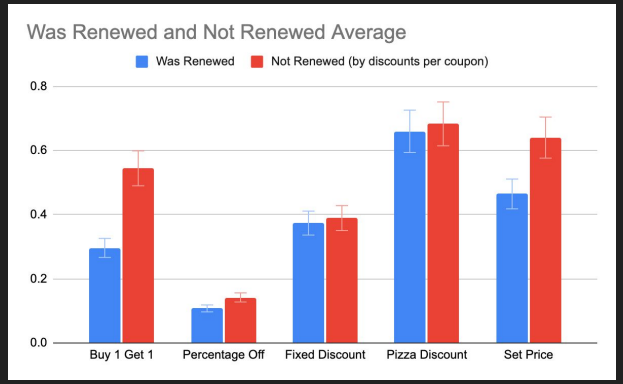
- Additionally, though we attempted to standardize the pizza into sizes, the way they are presented differs from store to store and was very difficult to standardize, leaving room for error.
  - ex. Some stores had sizes beyond extra-large like mountain/family size. Though they were different sizes, we had to group these all in the same category.

Finding Trends in Coupon Usage: Conclusion

- The pizza sizes and corresponding discount amounts did not appear to influence the success of the coupons when measured in coupons used per week.

- Advertisers should therefore use coupons with a discount of at most $2-$3 to maximize margins while receiving the same increase in traffic.

Method 2: Finding Trends in Coupon Renewal



Finding Trends in Coupon Renewal: Objective

Objective:

Determine how the type of discount offered correlates to renewal rates

Factors tested:

- Type of discount used
- Whether the coupon was renewed and given a testimonial

```
Select CONCAT('https://skynet.indoormedi
From Testimonials t
JOIN FileAssets a on t.ID = a.EntityID
WHERE a.EntityType = 25
ORDER BY t.ID DESC
```

Figure 6: SQL Query

Finding Trends in Coupon Renewal: Methodology

- Used SQL Query to extract data from the database image of coupon used and whether or not the coupon was renewed
- Analyzed each individual coupon and recorded the types and amounts of discounts offered
  - Discount categories were: Buy & Get, Fixed Discount, % Discount, Pizza Discount, and Set Price
- Created a graph based on recorded data to analyze trends.

Buy 1 Get 1	Percentage Off	Fixed Discount	Pizza Discount	Set Price	Image #	Was renewed / ga
1	0	0	3	0	1	Yes
0	0	0	3	0	2	Yes
0	0	0	1	0	3	Yes
1	0	1	0	0	4	Yes
0	0	0	1	0	5	Yes
0	0	0	0	2	6	Yes

Figure 7: example rows of spreadsheet

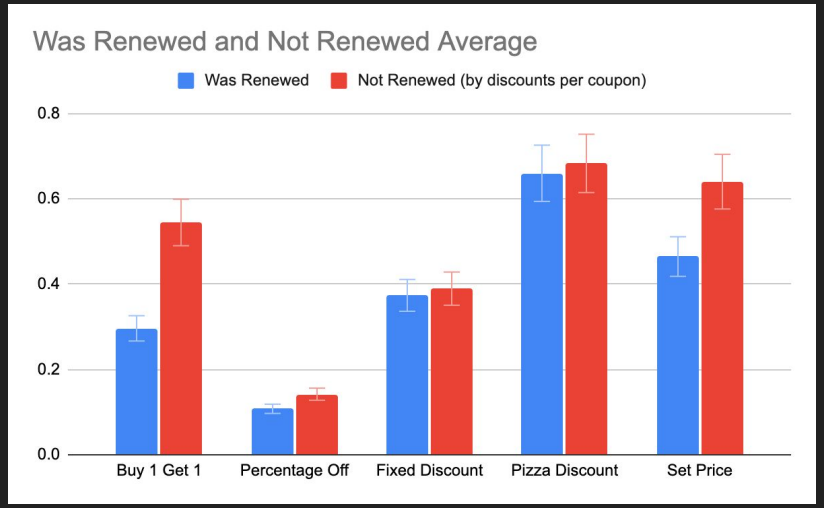
Finding Trends in Coupon Renewal: Example Coupon

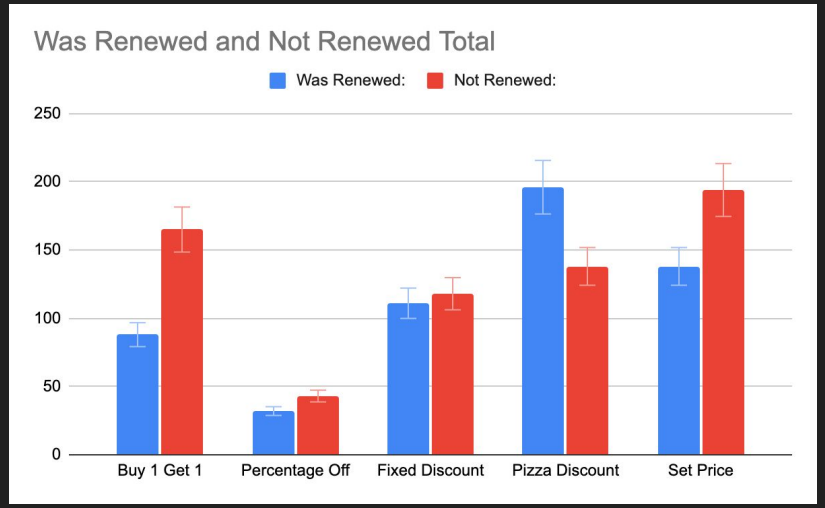
- In the Joe's Pizza & Pasta Coupon, there is 1 pizza discount and 1 percentage discount, so a 1 was put in the pizza discount and the percentage off columns
- Multiple slots were given in case the coupon offered multiple deals.

Figures 8, 9: Example coupon and corresponding table entry

Buy 1 Get 1	Percentage Off	Fixed Discount	Pizza Discount	Set Price	Image #	Was renewed / gave testimonial	
0	1	0	1	0	13	Yes	https://skynet.indoormedia.com/Coupons/Image/1wm7ToruBwiwQkhMme6gwcQi914icVMJO

Finding Trends in Coupon Renewal: Findings





Figures 10, 11: bar graphs of by coupon type, for renewed coupons w/ testimonials and unrenewed coupons

- Buy 1 Get 1 and Set Price offers are more commonly found in coupons that are not renewed
- Other discounts have similar appearance rates in both groups of coupons

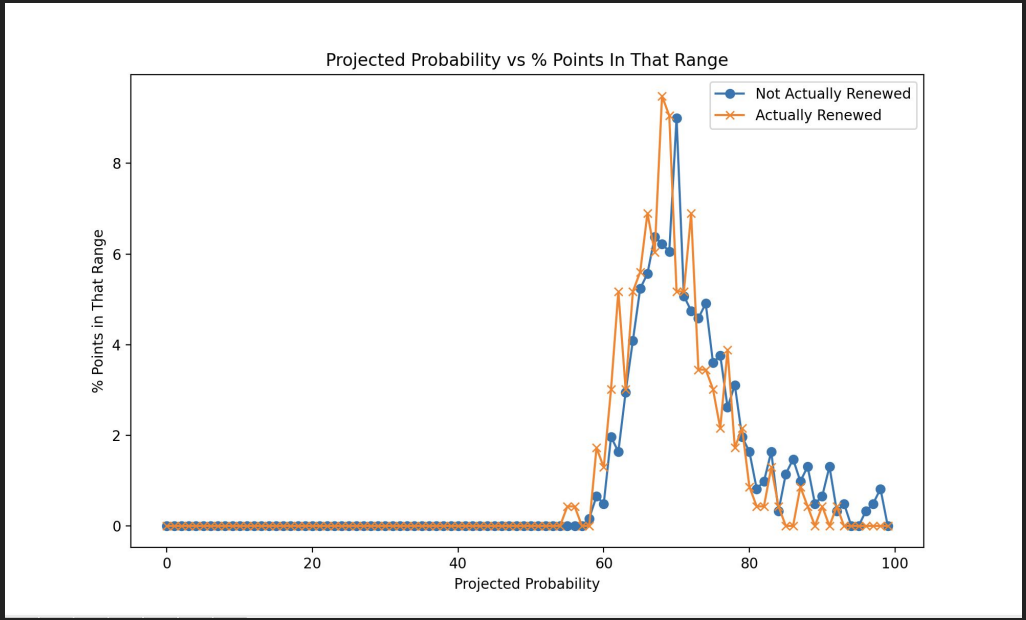
Finding Trends in Coupon Renewal: Conclusion

- According to the coupons analyzed, Buy 1 Get 1 discounts and Set Price discounts appear more in non-renewed coupons than renewed coupons.

- Designers should therefore encourage coupons with percentage off, fixed discount, and pizza discount coupons to support renewal, customer satisfaction*.

* further research necessary to determine causal relationship

Method #3: Machine Learning with Logistic Regression Analysis



Machine Learning With Logistic Regression Analysis: Objective

Objective:

Determine factors that encourage contract renewals among pizza clients.

Factors tested:

- Demographics in 3 mile radius around grocery store where advertised according to 2023 census data (total population, household income, gender, governmental support, age)
- Ad run length
- Total contract value

TotalAdPrice	RunLength	TotPop	MedianAge	pctMales	NumHHFoodStmp	NumHHRetInc	MedianHHInc	WasRenewed
3000.00	4	20287	46.2	47.6	670	2484	73454.00	0
2450.00	2	80473	35.7	49.5	2844	5176	74423.00	1
4615.00	4	93123	42.8	47.2	2178	6789	110205.00	0
4025.00	4	8910	49.9	47.2	612	1196	50291.00	1
1575.00	2	77797	40.3	50.2	1176	7766	93557.00	0
2968.00	4	24752	44.4	48.4	57	2329	175776.00	1
3816.00	4	16891	42.7	48.7	302	1312	129069.00	0
2421.00	4	89315	42	49	1036	8445	122565.00	0
4176.00	4	47581	45.2	49.7	610	5855	99870.00	0
3374.01	4	27657	40.4	49.9	1093	2817	50336.00	0
3726.60	4	142693	36	50.2	2127	8960	127418.00	0
2268.00	4	28320	35.8	51.2	1052	1901	60810.00	0
9000.00	12	15643	41.7	48.7	501	1644	85712.00	0
1698.00	2	156218	44.4	48.8	3986	12863	94075.00	0
2160.00	4	115962	42.7	49.4	3748	8541	83329.00	0
3550.00	4	13678	38.8	48.4	139	1227	100528.00	0
2556.00	4	34057	37.1	48.8	1548	2926	61638.00	0
3082.50	4	23747	43	47.4	1063	2825	52854.00	0

Figure 12: top 18 rows of table returned from SQL query

Machine Learning With Logistic Regression Analysis: Methodology

- Generated necessary SQL query to combine useful fields from StoreDemographics2023_3miles_EJS_TMP, sn_util_DougsLogAllProducts, ProductConfig_TapeAds, and Contracts
- Used Python “Pandas” and “NumPy” library to convert .xlsx file into array
- Created a logistic regression model using the “Sklearn” library designed to assign weights to each considered factor
- Used existence of renewal contract as a heuristic to measure ad success, therefore used as output of model (class 0: not renewed, class 1: renewed)
- Trained model on 80% of the data, tested on remaining 20% to determine model’s accuracy

```sql
select con.TotalAdPrice, prod.RunLength, census.TotPop, MedianAge, pctMales, NumHHFoodStmp, NumHHR
from sn_util_DougsLogAllProducts p
join StoreDemographics2023_3miles_EJS_TMP census on StoreName = name
join ProductConfig_TapeAds prod on prod.ID = p.ProductID
join Contracts con on con.ID = p.ContractID
where p.ProductTypeID = 0 AND (year(con.CreateDate) between 2021 and 2023) and p.CategoryID = 307
order by ReceivedDate DESC
```

Figure 13: SQL Query

Machine Learning With Logistic Regression Analysis: Findings

- Model successfully classifies product to renewal case with an accuracy of 72%
- Q: Is this model still useful / does it give actionable information?

```
              precision    recall  f1-score   support

           0       0.72      1.00      0.84       611
           1       0.00      0.00      0.00       232

    accuracy                           0.72       843
   macro avg       0.36      0.50      0.42       843
weighted avg       0.53      0.72      0.61       843

Weights [[-9.21311608e-05  6.42879947e-08 -2.33795043e-06 -1.69252277e-07
  -1.85020794e-07 -2.26909734e-06  1.39119441e-05 -4.27492757e-06]]
```

Figure 14: Classification report of model with weights for each factor

Machine Learning With Logistic Regression Analysis: Model Biases

- A: No
- Base rate of 72% of ads do not receive a renewal contract, identical to the model's predictive success rate
- Though the factors tested may or may not correlate, the high base rate overpowers any minor influences
- I.e. the model always guesses a resounding no

```
              precision    recall  f1-score   support

           0       0.72      1.00      0.84       611
           1       0.00      0.00      0.00       232

    accuracy                           0.72       843
   macro avg       0.36      0.50      0.42       843
weighted avg       0.53      0.72      0.61       843

Weights [[-9.21065984e-05  6.42925213e-08 -2.33167024e-06 -1.68829196e-07
  -1.83948137e-07 -2.43504197e-06  1.39112779e-05 -4.27766205e-06]]
Accuracy if 0 guessed every time: 72%
```

Figure 15: The precision of the model is identical to guessing against renewal every time

Machine Learning With Logistic Regression Analysis: Base Rate Fallacy, False Positive Paradox

- The dataset reveals that 70% of contracts from 2021-2023 are not renewed, thus establishing a base rate skewed towards class 0 (not renewed)
- Though factors may feature true correlations, a logistic regression model cannot draw reliable results due to the high “base rate” of contracts that are not renewed
- These shortcomings parallel the Base Rate Fallacy, which states that though certain factors might correlate to a specific result, a pre-existing and skewed base rate may ultimately weigh more due to the sheer likelihood of that result.

Machine Learning With Logistic Regression Analysis: Base Rate Fallacy, False Positive Paradox

- While we strive to create a model that acts on featural evidence, the best conclusion in this context is to assume the result that is by far more likely, irrespective of of demographic and product features.
- Though factors such as HHInc, Age, etc may have a role in influencing a contracts renewal rate, the renewal rate is so low that each contract, no matter its qualities, likely to not be renewed.

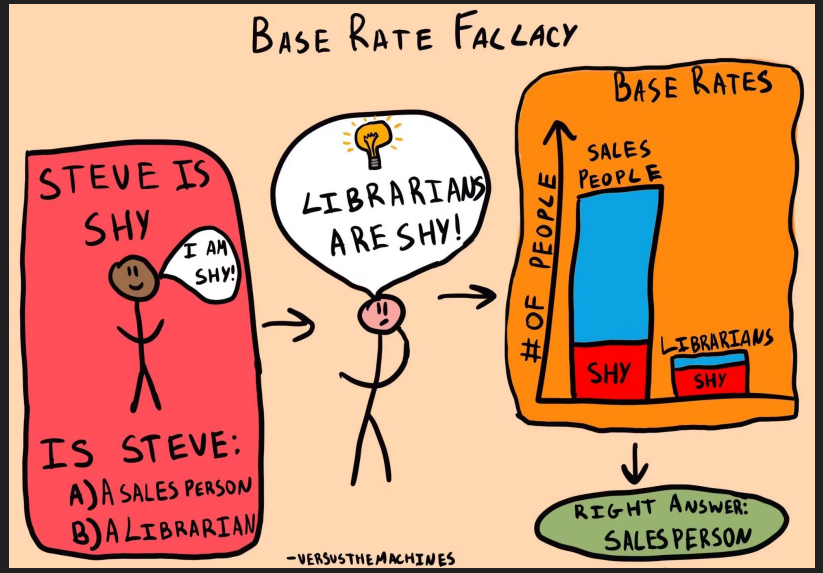


Figure 16: Visual Illustration of the Base Rate Fallacy (source: @versusthemachines)

Machine Learning With Logistic Regression Analysis: Amending Model

- Data was ‘balanced’ with the class_weight = “balanced” command, normalizing the model for less frequent null result
- Pro: far better identification of positive result
- Con: worse overall accuracy

```
              precision    recall  f1-score   support

           0       0.79      0.50      0.61       611
           1       0.33      0.65      0.44       232

    accuracy                           0.54       843
   macro avg       0.56      0.57      0.52       843
weighted avg       0.66      0.54      0.56       843

Weights [[-6.99865721e-05  4.05207509e-07 -5.89105621e-06  2.43969913e-06
   3.25689899e-06  6.74514151e-05  4.34889496e-05  3.89450781e-06]]
```

Figure 17: Balanced classification report and weights

Machine Learning With Logistic Regression Analysis: Possible Inaccuracies

- This model assumes that each factor is linearly correlated with renewal rate, and fails to consider that some factors may have a quadratic or other non-linear relationship with the output.
  - EX: Extremely low-income households consume from restaurants less and therefore businesses in those areas may benefit less from coupon advertising, but middle-class and upper-class people both use coupons an equal amount. Coupon usage (and success) would increase as demographics progress from low-class to middle-class, but then level out as demographics move beyond middle-class.
- The model assumes each factor to be independent, but in reality many factors may be correlated with each other.
  - EX: Median household income and food stamps are correlated and median age and retirement income are correlated as well.

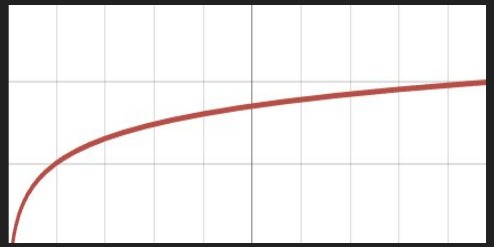


Figure 18: Example of a nonlinear relationship (from Desmos)

Machine Learning With Logistic Regression Analysis: Probabilities

- Because the probability that any given case would be 0 was higher than 50%, the model guessed class 0 a large portion of the time
- However, the probability that the test case would have a renewal rate of 0 varied across different cases. Thus looking at not just the predicted result but also the probability of renewal rates for any given case allows us to further measure the accuracy of the model
- Realizing this impossible threshold, we analyzed the probabilities the model provided to find possible insights and true correlations

```python
for num in range(50):
    X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.2, random_state=num)
    probabilities = model.predict_proba(X_test)
    ones_count = 0
    ones_total_probability = 0
    zeroes_count = 0
    zeroes_total_probability = 0
    for i in range(len(X_test)):
        if y_test[i] == 0:
            zeroes_count += 1
            zeroes_total_probability += probabilities[i]
        else:
            ones_count += 1
            ones_total_probability += probabilities[i]
    print("Average probability of result 0 where actual result is 0: " + str(zeroes_total_probability / zeroes_count))
    print("Average probability of result 0 where actual result is 1: " + str(ones_total_probability / ones_count))
    total_difference += (zeroes_total_probability / zeroes_count)[0] - (ones_total_probability / ones_count)[0]
# Print the probabilities and the actual results
average_difference = total_difference / 50
print("Average Difference After 50 trials: " + str(average_difference))
```

Figure 19: Python code to find average difference in probability between renewed unrenewed contracts

Machine Learning With Logistic Regression Analysis: Probabilities

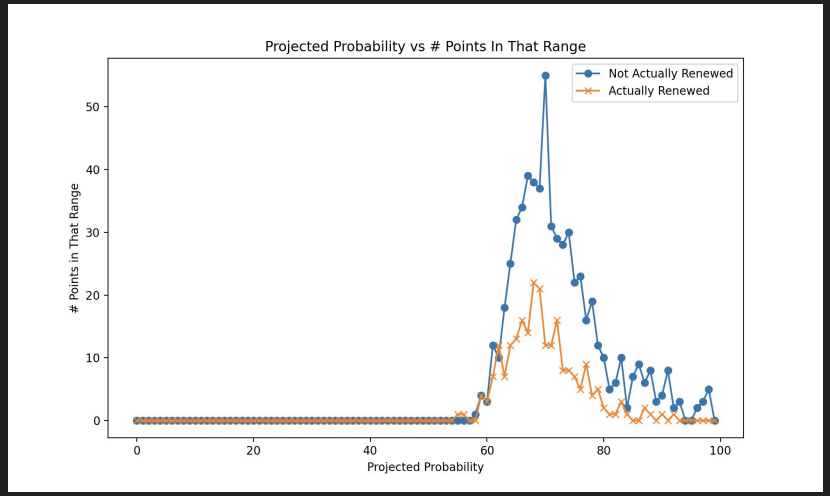


Figure 20: # of points in projected probability ranges

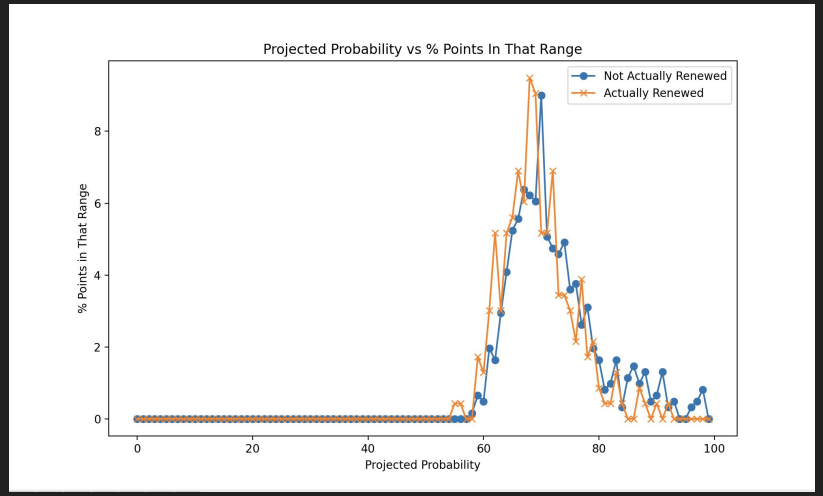


Figure 21: % of points in projected probability ranges

Above is the projected probability vs. the points in that range. Though they are close, the renewed class has a visibly less projected probability than the not-renewed class. This demonstrates that the algorithm is detecting a difference, though minimal, between the classes.

Machine Learning With Logistic Regression Analysis: Actionable Insights

Since the model is 3.2 percentage points better at identifying contracts that were renewed, the model's factors have true though minor correlations with renewal rate.

```python
# Get the weights (coefficients) of the model
weights = model.coef_
print('Weights:', weights)

correct_count = 0
total_elements = 0
for element in y_test:
    if element == 0:
        correct_count += 1

    total_elements += 1

print("Accuracy if 0 guessed every time: " + str((correct_count / total_elements)))
```

```
Average probability of result 0 where actual result is 0: [0.72627359 0.27372641]
Average probability of result 0 where actual result is 1: [0.68756638 0.31243362]
Average Difference After 50 trials: 0.031552244509299915
```

Figures 22, 23: The data on the past slide visualized. There is a minimal difference (3.2 Percentage Points) between the algorithms calculated probabilities for class 0 and 1.

Machine Learning With Logistic Regression Analysis: Final Insights

Looking at the signs and values for the weights as shown in Figure 2, we have reached the following conclusions about the discussed factors:

1. Negatively Correlated: Higher ad prices, total population, median age, percentage of males, number of households on food stamps, and median household income
2. Positively Correlated: Run length, retirement income are positively correlated with contract renewal
3. Ignore all of these correlations before making a decision that contradicts with common sense

TotalAdPrice	RunLength	TotPop	MedianAge	pctMales	NumHHFoodStmp	NumHHRetInc	MedianHHInc	WasRenewed
3000.00	4	20287	46.2	47.6	670	2484	73454.00	0
2450.00	2	80473	35.7	49.5	2844	5176	74423.00	1
4615.00	4	93123	42.8	47.2	2178	6789	110205.00	0
4025.00	4	8910	49.9	47.2	612	1196	50291.00	1

```
              precision    recall  f1-score   support

           0       0.72      1.00      0.84       611
           1       0.00      0.00      0.00       232

    accuracy                           0.72       843
   macro avg       0.36      0.50      0.42       843
weighted avg       0.53      0.72      0.61       843

Weights [[-9.21311608e-05  6.42879947e-08 -2.33795043e-06 -1.69252277e-07
  -1.85020794e-07 -2.26909734e-06  1.39119441e-05 -4.27492757e-06]]
```

Figures 24,25: SQL Data of factors table, corresponding weights assigned to factors respectively

Machine Learning With Logistic Regression Analysis: Conclusion

The model was unable to predict whether or not a given ad would be renewed accurately. However, since the model was able to detect a difference between the non-renewed and renewed contracts' probabilities of being renewed, there therefore are minor correlations between selected factors and renewal rates.